Day 083 | Unit 06 | Teacher plan

Understand and Use Ratios and Rates

Understand Rates and Unit Rates

Standards: 6.RP.A.2, 6.RP.A.3b

Learning target: I can find a unit rate and explain its units.

enVision Topic 5: Understand and Use Ratios and Rates | Spectrum companion Chapter 3

0–5 minutes / Launch

Read the target. Ask for an estimate or a sketch before calculating. Connect the new idea to the previous lesson.

5–17 / Model

Use Worked Example 1, then Example 2. Ask students to name the next step and explain a choice. A unit rate compares an amount with one unit of another quantity. Divide the first quantity by the second. Write the units, such as dollars per notebook, so you know what the quotient means.

17–42 / Practice

Assign questions 1–10 in order. Use 1–3 for foundations, 4–7 for application, and 8–10 for state-format response and reasoning. The example button provides a separate question with a worked method.

42–50 / Discuss and check

Compare representations. Review a selected answer and a plausible error. Use question 10 as an exit discussion; ask students to check units, scale, or substitution.

Support and extension

Offer number lines, fraction strips, grid paper, or a table template as appropriate. Read multi-part directions one part at a time. For extension, ask for another representation or a counterexample.

Companion and state alignment

Use the matching skill in the edition you own. No unverified publisher page numbers are assigned. AASA references below identify related public sample items by standard and format; all course questions are original.

AASA sample item references: 2, 21

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01 / Learn

A unit rate compares an amount with one unit of another quantity. Divide the first quantity by the second. Write the units, such as dollars per notebook, so you know what the quotient means.

Worked example 1

5 identical notebooks cost $10. Find the cost per notebook.

1. A unit rate compares an amount with one unit of another quantity.

2. Divide the first quantity by the second.

3. Unit rate = 10 ÷ 5 = 2 dollars per notebook.

4. State the result in the form requested: $2 per notebook.

Answer: $2 per notebook

Worked example 2

6 notebooks cost $42. Select all statements at the same price per notebook.
A. 5 notebooks cost $36.
B. 5 notebooks cost $35.
C. Each notebook costs $252.
D. Each notebook costs $7.

1. A unit rate compares an amount with one unit of another quantity.

2. Cost per notebook = 42 ÷ 6 = 7.

3. Multiply $7 by the chosen notebook count.

Answer: B, D - 5 notebooks cost $35.; Each notebook costs $7.

Day 083 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

6 identical notebooks cost $18. Find the cost per notebook.

02 / Constructed response

4 identical notebooks cost $12. Find the cost per notebook.

03 / Constructed response

7 identical notebooks cost $42. Find the cost per notebook.

04 / Constructed response

7 identical notebooks cost $21. Find the cost per notebook.

05 / Constructed response

2 identical notebooks cost $12. Find the cost per notebook.

06 / Constructed response

6 identical notebooks cost $42. Find the cost per notebook.

07 / Constructed response

7 identical notebooks cost $14. Find the cost per notebook and the cost of 10 notebooks at this rate.

08 / Multiple select

6 notebooks cost $12. Select all statements at the same price per notebook.
A. Each notebook costs $2.
B. 4 notebooks cost $8.
C. 4 notebooks cost $9.
D. Each notebook costs $72.

09 / Multiple select

5 notebooks cost $10. Select all statements at the same price per notebook.
A. 5 notebooks cost $10.
B. Each notebook costs $50.
C. Each notebook costs $2.
D. 5 notebooks cost $11.

10 / Constructed response

6 identical notebooks cost $15. Find the cost per notebook and the cost of 9 notebooks at this rate.

Day 083 | Unit 06 | Answer key

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Learning target: I can find a unit rate and explain its units.

1. $3 per notebook

Unit rate = 18 ÷ 6 = 3 dollars per notebook.

6.RP.A.2, 6.RP.A.3b | Constructed response

2. $3 per notebook

Unit rate = 12 ÷ 4 = 3 dollars per notebook.

6.RP.A.2, 6.RP.A.3b | Constructed response

3. $6 per notebook

Unit rate = 42 ÷ 7 = 6 dollars per notebook.

6.RP.A.2, 6.RP.A.3b | Constructed response

4. $3 per notebook

Unit rate = 21 ÷ 7 = 3 dollars per notebook.

6.RP.A.2, 6.RP.A.3b | Constructed response

5. $6 per notebook

Unit rate = 12 ÷ 2 = 6 dollars per notebook.

6.RP.A.2, 6.RP.A.3b | Constructed response

6. $7 per notebook

Unit rate = 42 ÷ 6 = 7 dollars per notebook.

6.RP.A.2, 6.RP.A.3b | Constructed response

7. $2 per notebook; $20

Unit rate = 14 ÷ 7 = 2 dollars per notebook.

6.RP.A.2, 6.RP.A.3b | Constructed response

8. A, B - Each notebook costs $2.; 4 notebooks cost $8.

Cost per notebook = 12 ÷ 6 = 2. Multiply $2 by the chosen notebook count.

6.RP.A.2, 6.RP.A.3b | Multiple select

9. A, C - 5 notebooks cost $10.; Each notebook costs $2.

Cost per notebook = 10 ÷ 5 = 2. Multiply $2 by the chosen notebook count.

6.RP.A.2, 6.RP.A.3b | Multiple select

10. $2.5 per notebook; $22.5

Unit rate = 15 ÷ 6 = 2.5 dollars per notebook.

6.RP.A.2, 6.RP.A.3b | Constructed response